Customer Problem

Using AWS · GITHUB · TRELLO

Customer

A fast-growing startup has launched a new web application, but users are experiencing slow page load times, inconsistent availability during peak traffic, and concerns about the security of the application. The company also lacks a structured deployment process, making collaboration between developers difficult and increasing the risk of deployment errors. They need a secure, highly available, and scalable cloud solution that can be deployed within a limited budget while supporting future growth.

PROJECT

End-to-End Cloud Solution Design & Deployment

Objective

Design, architect, and deploy a secure, scalable web application on AWS using CloudFront, ALB, and related services. Interns will collaborate using GitHub for version control and Trello for project management, building real-world cloud deployment skills within the AWS Free Tier.

CloudFront

ALB

EC2/S3

GitHub

Trello

Phased Milestones

Phase 5: Intern Presentation, Project Showcase

Challenge: Present your completed cloud project, explain architectural decisions, and demo the live deployment

What to Present

- Problem statement: What problem does your web app solve?
- Architecture walkthrough: Explain CloudFront → ALB → EC2 flow
- Security decisions: How is the deployment secured end-to-end?
- CI/CD demo: Show a live GitHub Actions pipeline run
- Trello board review: Walk through completed sprints and task history
- Live demo: Access the deployed app via the CloudFront URL (HTTPS)
- Challenges faced: Discuss blockers encountered and how they were resolved
- Lessons learned and what you'd do differently next time

Deliverables

- GitHub repo with all source code
- Completed Trello board (screenshot)
- Architecture diagram
- Live HTTPS URL (CloudFront)
- README with setup guide
- Presentation slides (this deck)
- Recorded demo video (optional)

Expected Outcome Interface Example

Secure Web App, Deployed on AWS

🔒 https://d1abc123.cloudfront.net (HTTPS · CloudFront CDN)

☁ **MyCloudApp | Home About Dashboard Contact**

Welcome to Your Deployed Web App

Served securely via CloudFront CDN · Routed by Application Load Balancer · Hosted on EC2

HTTPS Secured
SSL via ACM + CloudFront

Load Balanced
ALB Health Checks Active

CDN Enabled
Global CloudFront Edge

CI/CD Live
GitHub Actions Pipeline

Links to Resources and Documentation

Documentation for all tools used in this project

Amazon CloudFront
https://docs.aws.amazon.com/cloudfront

Application Load Balancer
https://docs.aws.amazon.com/elasticloadbalancing

Amazon EC2
https://docs.aws.amazon.com/ec2

AWS Certificate Manager
https://docs.aws.amazon.com/acm

AWS IAM
https://docs.aws.amazon.com/iam

Amazon S3
https://docs.aws.amazon.com/s3

GitHub Actions
https://docs.github.com/en/actions

Trello Guide
https://trello.com/guide

AWS Budgets
https://docs.aws.amazon.com/awsaccountbilling/latest/aboutv2/budgets-managing-costs.html

AWS Free Tier
https://aws.amazon.com/free

Remember: The goal is not just to build the application, but to understand the WHY behind each architectural decision. Focus on learning principles that will apply to future projects!